

PORT WASHINGTON, WI, USA

WMO: 997736

Lat: 43.3881N

Lon: 87.8667W

Elev: 591

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	1.7	N/A	N/A	N/A	N/A	N/A	N/A	26.1	27.8	24.0	30.3	9.8	280	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	86.8	N/A	83.0	N/A	79.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.2	250

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.1	18.8	16.6	-7.2	94.5	4.9	4.3	-10.8	97.6	-13.7	100.1	-16.5	102.6	-20.0	105.7		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	21.5	22.9	33.4	43.6	53.6	63.3	70.1	69.8	62.8	51.0	39.0	28.8
	DBStd	18.74	11.64	10.29	9.12	6.42	6.32	6.75	6.21	5.23	6.28	6.99	8.91	9.67
	HDD50	3477	883	760	516	213	32	0	0	0	1	76	340	656
	HDD65	7105	1348	1180	978	643	357	111	22	17	113	436	779	1121
	CDD50	2305	0	0	2	20	143	401	622	615	384	106	12	0
	CDD65	460	0	0	0	0	3	61	180	168	47	1	0	0
	CDH74	2447	0	0	0	7	21	371	1101	751	183	13	0	0
	CDH80	678	0	0	0	1	2	90	366	175	42	2	0	0
Wind	WSAvg	7.5	8.7	8.9	8.4	8.7	6.9	5.9	5.7	5.9	6.5	7.6	7.9	8.4
Precipitation	PrecAvg	33.7	1.7	1.4	1.9	3.5	3.9	4.6	3.8	3.7	2.7	2.4	2.1	1.7
	PrecMax	42.4	4.6	3.1	4.4	6.2	11.7	13.2	7.6	8.2	6.9	5.4	5.3	3.9
	PrecMin	27.5	0.3	0.1	0.4	0.9	1.0	1.5	1.9	2.0	0.0	0.7	0.3	0.3
	PrecStd	3.9	1.0	0.9	1.1	1.4	2.3	2.8	1.7	1.6	1.5	1.4	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	50.1	61.6	70.3	80.4	87.3	95.4	90.4	85.2	74.9	64.0	53.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	44.1	41.5	54.4	61.9	72.7	82.6	88.4	85.0	78.5	67.0	57.5	46.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	40.3	38.3	50.0	56.3	68.0	78.4	84.0	80.5	75.0	64.0	52.9	44.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	37.4	35.9	45.4	52.2	63.6	74.0	79.9	77.8	71.9	61.1	50.1	41.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.9	12.2	11.5	13.0	13.1	14.0	13.3	13.2	12.9	11.6	11.4
		MCDBR	15.2	14.3	17.8	19.1	20.7	21.0	20.4	17.9	17.5	17.2	14.8	13.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.288	0.316	0.342	0.389	0.412	0.431	0.424	0.417	0.390	0.346	0.309	0.287	
	taud	2.429	2.336	2.374	2.284	2.274	2.269	2.292	2.305	2.365	2.476	2.518	2.475	
	Ebn at Noon	270	278	285	278	273	267	268	266	265	265	260	260	
	Edh at Noon	23	31	34	40	42	42	41	39	34	27	22	21	
All-Sky Solar Radiation	RadAvg	469	768	1142	1446	1699	1907	1953	1673	1347	857	534	374	
	RadStd	65	87	114	132	141	132	88	106	111	97	47	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page